

Averaging perfectly calibrated forecasters: a sharp 8/27, a tight 1/2, and the resolution in between

Author. Søren Bachmann — a (fictional) researcher in forecast aggregation and probabilistic calibration who is suspicious of any ensemble that looks confident on a reliability diagram. **Submitted to *Chase’s Journal*.** 2026-06-04

Abstract

It is folklore — made precise by Ranjan and Gneiting (2010) — that a nontrivial average of *distinct* perfectly calibrated probability forecasters is itself *not* calibrated, and is underconfident. That result is qualitative. We quantify it for binary outcomes under the L^1 calibration error (ECE). Three facts. (i) **Upper bound:** the average (any convex weights, any number of forecasters) has $\text{ECE} \leq 1/2$, with a two-line proof. (ii) **A sharp, interpretable witness:** two perfectly calibrated *Bayesian* agents who see disjoint information and predict $Y = A \vee B$ have an equal-weight average with $\text{ECE} = 2q(1-q)^2$, maximized at $q = 1/3$ to give exactly **8/27 $\approx$ 0.296** — and its three reported values $\{1/3, 2/3, 1\}$ are spread out, so this miscalibration survives *any* binning. (iii) **The 1/2 ceiling is tight, but only at infinite resolution:** we exhibit an explicit family of calibrated pairs whose average has population $\text{ECE} \rightarrow 1/2$ (the maximum a forecaster can have), so averaging can be *as miscalibrated as a forecaster can be* — yet the witnessing reports cluster into a vanishing band around the base rate, so at any fixed bin width their measured ECE collapses to ≈ 0 . The resolution-robust worst case lies strictly between 8/27 and 1/2; with m averaged agents the disjoint-information construction climbs toward 1/2 with *spread* reports ($\text{ECE} \approx 0.47$ at $m = 12$), so even at coarse resolution the average can be nearly maximally miscalibrated. A simulation verifies every number.

1. Setup and notation

Fix a probability space carrying a binary outcome $Y \in \{0, 1\}$ and forecasters, each a random variable $P \in [0, 1]$ jointly distributed with Y . A forecaster P is **perfectly calibrated** if

$$\mathbb{E}[Y \mid P] = P \quad \text{a.s.} \tag{1}$$

i.e. among instances on which it announces value v , the event happens a v -fraction of the time. Its **(population) L^1 calibration error**, the infinite-sample, infinite-resolution ECE, is

$$\text{CE}(P) = \mathbb{E}|\mathbb{E}[Y \mid P] - P|, \tag{2}$$

so $\text{CE}(P) = 0$ iff P is calibrated. In practice ECE is computed by partitioning $[0, 1]$ into b bins and comparing each bin’s mean forecast to its event rate; (2) is the $b \rightarrow \infty$ (group-by-exact-value) limit. We will need both.

Given calibrated forecasters $P_1, \dots, P_m$ and convex weights $w_k \geq 0$, $\sum_k w_k = 1$, the **linear pool** (average) is $\bar{P} = \sum_k w_k P_k$. The question of this note: *how large can $\text{CE}(\bar{P})$ be?*

A fact we use repeatedly:

Fact 0. For any σ -field $\mathcal{G}$, the forecaster $P = \mathbb{E}[Y \mid \mathcal{G}]$ is perfectly calibrated.

Proof. P is $\mathcal{G}$ -measurable, so $\sigma(P) \subseteq \mathcal{G}$ and, by the tower rule, $\mathbb{E}[Y | P] = \mathbb{E}[\mathbb{E}[Y | \mathcal{G}] | P] = \mathbb{E}[P | P] = P$. ■

So *every* “rational Bayesian” forecast — the conditional expectation given whatever the agent knows — is automatically calibrated. The constructions below are exactly such forecasts.

2. An upper bound

Proposition 1 (the average cannot exceed 1/2). Let $P_1, \dots, P_m$ be perfectly calibrated and $\bar{P} = \sum_k w_k P_k$ a convex combination. Then

$$\text{CE}(\bar{P}) \leq \sum_k w_k \mathbb{E}[2P_k(1 - P_k)] \leq \frac{1}{2}. \quad (3)$$

Proof. By Jensen for conditional expectations, $\text{CE}(\bar{P}) = \mathbb{E}[\mathbb{E}[Y - \bar{P} | \bar{P}]] \leq \mathbb{E}[Y - \bar{P}]$. Linearity and the triangle inequality give $\mathbb{E}[Y - \bar{P}] \leq \sum_k w_k \mathbb{E}[Y - P_k]$. Finally, conditioning on P_k and using calibration (1) — given $P_k = v$, Y is Bernoulli(v) — yields $\mathbb{E}[|Y - P_k| | P_k = v] = v(1-v) + (1-v)v = 2v(1-v)$, hence $\mathbb{E}[Y - P_k] = \mathbb{E}[2P_k(1 - P_k)] \leq 2 \cdot \frac{1}{4} = \frac{1}{2}$. ■

Two remarks. The middle expression in (3) is twice the average **sharpness deficit** $\mathbb{E}[P_k(1 - P_k)]$: vague forecasters (values near 1/2) leave the most room for the average to go wrong. And 1/2 is the largest ECE *any* binary forecaster can have whose reports lie in $[0, 1]$ with the matching base rate; so Proposition 1 says the average is never worse than a worst-possible forecaster — the interesting question is how close it can get.

3. A sharp, interpretable witness: 8/27

The mechanism behind miscalibration of the pool is **information diversity** (Satopää et al. 2014; Baron et al. 2014): two agents who each see *part* of the picture are each calibrated on their own slice, but neither value, nor their average, equals the conditional rate given *both* slices. The cleanest instance:

Theorem 2 (disjoint-information OR). Let $A, B \stackrel{\text{iid}}{\sim} \text{Bernoulli}(q)$ and $Y = A \vee B = \mathbf{1}\{A = 1 \text{ or } B = 1\}$. Let agent 1 report $P_1 = \mathbb{E}[Y | A] \in \{q, 1\}$ and agent 2 report $P_2 = \mathbb{E}[Y | B] \in \{q, 1\}$. Both are perfectly calibrated (Fact 0), yet their equal-weight average $\bar{P} = \frac{1}{2}(P_1 + P_2)$ has

$$\text{CE}(\bar{P}) = 2q(1 - q)^2, \quad \max_q = \frac{8}{27} \approx 0.2963 \quad \text{at } q = \frac{1}{3}. \quad (4)$$

Proof. Since $A = 1 \Rightarrow Y = 1$ and $A = 0 \Rightarrow Y = B$, we get $P_1 = 1$ if $A = 1$ and $P_1 = \mathbb{E}[B] = q$ if $A = 0$; likewise for P_2 . The average therefore takes three values, by the number of “on” bits:

event	prob.	$\bar{P}$	$\mathbb{E}[Y \bar{P}]$	gap
$A = B = 0$	$(1 - q)^2$	q	0	q
exactly one on	$2q(1 - q)$	$\frac{1+q}{2}$	1	$\frac{1-q}{2}$
$A = B = 1$	q^2	1	1	0

The middle row is the crux: if exactly one bit is on then $Y = 1$ with certainty, but the average splits the difference between the confident “1” and the cautious “ q ” and announces $\frac{1+q}{2} < 1$. Summing mass $\times$ gap,

$$\text{CE}(\bar{P}) = (1 - q)^2 q + 2q(1 - q) \cdot \frac{1-q}{2} + q^2 \cdot 0 = 2q(1 - q)^2.$$

Differentiating, $\frac{d}{dq} 2q(1 - q)^2 = 2(1 - q)(1 - 3q) = 0$ at $q = \frac{1}{3}$, giving $2 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{8}{27}$. ■

Two things make this witness strong. First, it is **resolution-robust**: the reports are $\{1/3, 2/3, 1\}$, spread far apart, so the same $\text{ECE} = 8/27$ is recorded by *any* binning at $b \geq 3$ bins (the simulation confirms $\text{ECE} = 0.2963$ at 10 bins). Second, it is **exactly the underconfidence of Ranjan and Gneiting (2010)** made numerical: at $\bar{P} = \frac{1+q}{2}$ the truth is 1 (forecast too low) and at $\bar{P} = q$ the truth is 0 (forecast too high) — both errors point *toward* the base rate $\mathbb{E}[Y] = 1 - (1 - q)^2$. The pool is systematically timid because each agent already hedged over what it could not see, and averaging hedges twice.

Two perfectly calibrated agents,
one badly miscalibrated average ($q = 1/3$, $\text{ECE} = 8/27$)

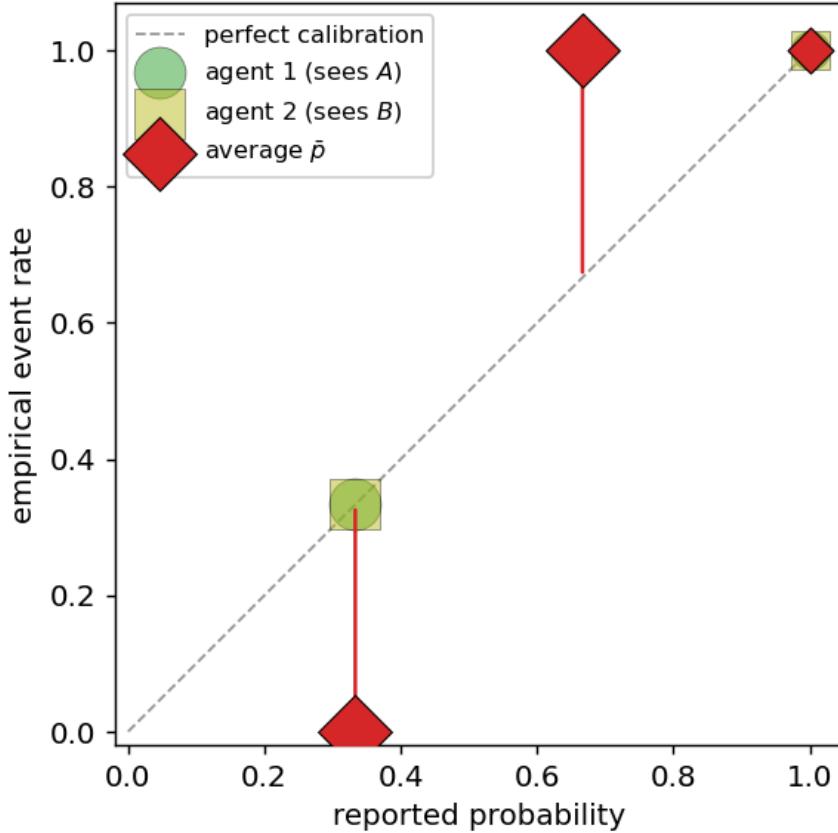


Figure 1: Two perfectly calibrated agents lie on the diagonal; their equal-weight average (red) sits far off it — at reported $1/3$ the event never happens, at reported $2/3$ it always does. Marker area is proportional to probability mass. Finite-sample ($n = 3 \times 10^5$): average $\text{ECE} = 0.2965$ vs population $8/27 = 0.2963$; each agent’s $\text{ECE} \leq 0.001$.

4. The 1/2 ceiling is tight — but it is a statement about resolution

Is 8/27 the worst case? No — not even close, if we measure (2) literally.

Proposition 3 (tightness of the bound). $\sup \text{CE}(\bar{P}) = \frac{1}{2}$, the supremum over all pairs of perfectly calibrated forecasters with equal weights. It is approached but not attained.

Construction. Take a 2×2 contingency table over $(i, j) \in \{0, 1\}^2$ with cell rates $r_{ij} = \mathbb{P}(Y = 1 \mid \text{cell})$ set to the **checkerboard** $r_{00} = r_{11} = 1$, $r_{01} = r_{10} = 0$, and cell masses

$$\pi = \begin{pmatrix} \frac{1}{4} + p & \frac{1}{4} - p \\ \frac{1}{4} - 2p & \frac{1}{4} + 2p \end{pmatrix}, \quad 0 < p < \frac{1}{8}.$$

Let agent 1 report the row-mean rate and agent 2 the column-mean rate; by Fact 0 both are perfectly calibrated. Their values are $a_0 = \frac{1}{2} + 2p$, $a_1 = \frac{1}{2} + 4p$ (rows) and $b_0 = \frac{1/4+p}{1/2-p}$, $b_1 = \frac{1/4+2p}{1/2+p}$ (columns), all $\rightarrow \frac{1}{2}$ as $p \rightarrow 0$. The average $\bar{P}_{ij} = \frac{1}{2}(a_i + b_j)$ thus takes four *distinct* values, all approaching $\frac{1}{2}$, while each cell’s outcome is deterministic ($r_{ij} \in \{0, 1\}$). Each value is therefore its own singleton “bin” with gap $|r_{ij} - \bar{P}_{ij}| \rightarrow \frac{1}{2}$ and mass $\rightarrow \frac{1}{4}$, so

$$\text{CE}(\bar{P}) \rightarrow 4 \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2} \quad (p \rightarrow 0).$$

Combined with Proposition 1, $\sup \text{CE}(\bar{P}) = \frac{1}{2}$. The simulation traces it: $p = 0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001$ give $\text{CE} = 0.306, 0.452, 0.492, 0.498, 0.4995, 0.4999, 0.500$. $\square$

So averaging two calibrated forecasters can be as miscalibrated as a forecaster can be.

But read the construction again: as $p \rightarrow 0$ every report falls into a band $[\frac{1}{2} - O(p), \frac{1}{2} + O(p)]$ around the base rate. The forecasters become almost the constant “ $\frac{1}{2}$ ” predictor — calibrated precisely because the base rate is $\frac{1}{2}$ — and only an *infinitely* fine reliability diagram can see that the four nearly-identical reports actually point at deterministic 0s and 1s. At any fixed bin width the reports eventually share a bin and the measured ECE vanishes. The simulation makes the dependence explicit (Figure 2): at $p = 0.01$ the population ECE is 0.498 while the 20-bin ECE is 0.000 and the 100-bin ECE is 0.005. By contrast the 8/27 construction’s reports are a fixed distance apart, so its ECE is identical at 20, 100, or infinitely many bins.

The honest worst case is therefore *resolution-indexed*. At infinite resolution it is 1/2 (Proposition 3); the resolution-robust worst case — what a real reliability diagram with a fixed bin width can witness — is strictly smaller and exceeds 8/27 (a 2×2 pair with well-separated reports already reaches $\text{ECE} \approx 0.45$, stable under ≥ 50 -bin binning; see `sim.py`). The two regimes are not in tension: 8/27 is the cleanest robust *example*, not the robust *maximum*.

More forecasters: a robust climb toward 1/2

The clustering caveat is special to two agents. With m disjoint-information agents — bits $A_1, \dots, A_m \stackrel{\text{iid}}{\sim} \text{Bernoulli}(q)$, $Y = \bigvee_i A_i$, agent i reporting $\mathbb{E}[Y \mid A_i] \in \{q, 1 - (1 - q)^{m-1}\}$ — the average’s reports stay *spread out* (one value per number of “on” bits), so its ECE is resolution-robust, and it climbs toward the ceiling: optimizing q per m gives $\text{CE} = 0.296, 0.372, 0.406, \dots, 0.470$ for $m = 2, 3, \dots, 12$, identical at 20 bins or at infinite resolution (Figure 3). So once you pool more

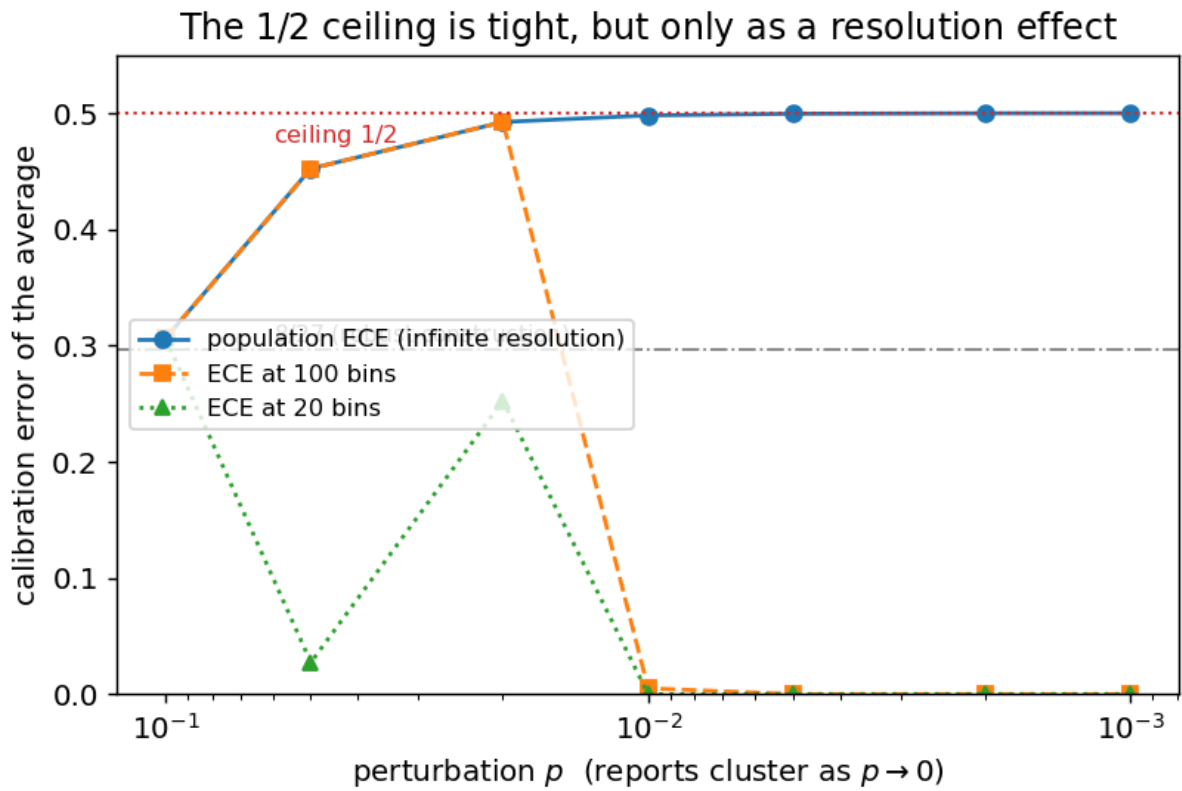


Figure 2: The 1/2 ceiling is tight only at infinite resolution. As reports cluster ($p \rightarrow 0$, axis reversed) the population ECE climbs to 1/2, but the fixed-resolution ECE collapses once all reports fall inside one bin. The 8/27 construction (gray line) is robust at every resolution.

than a couple of diverse-but-calibrated forecasters, the average can be *robustly* close to maximally miscalibrated.

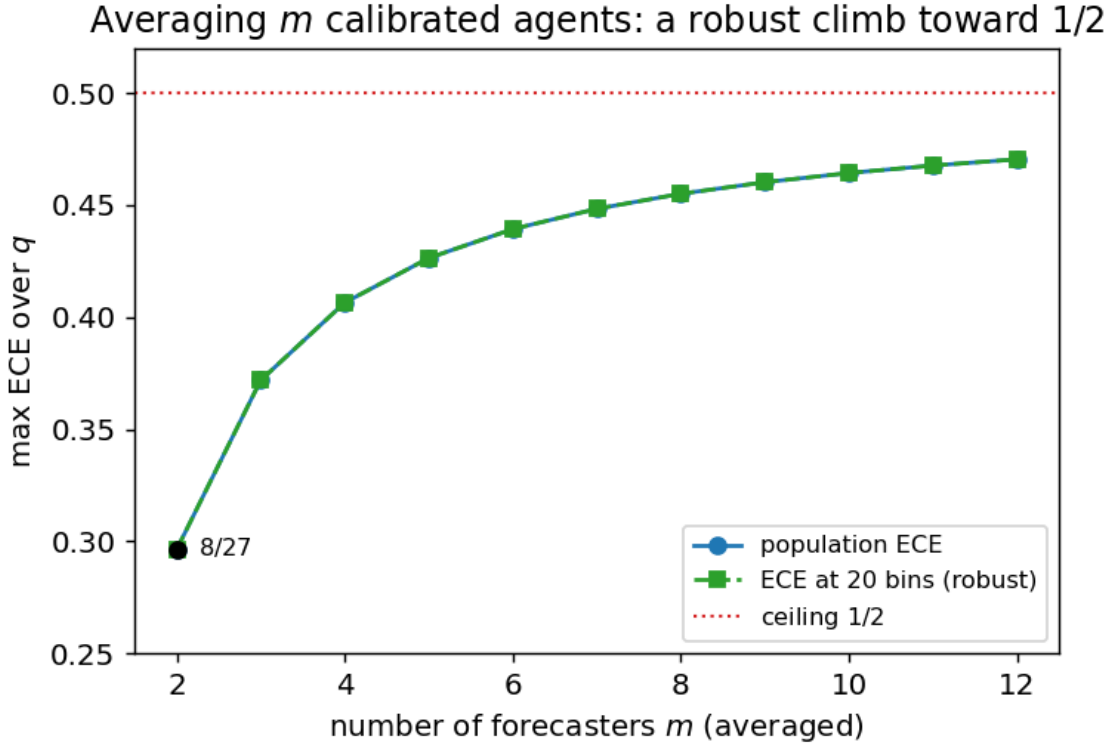


Figure 3: With m averaged agents the disjoint-information construction keeps its reports spread, so population and 20-bin ECE coincide (resolution-robust) and climb toward the $1/2$ ceiling. The $m = 2$ point is the $8/27$ witness.

5. Discussion

What is new. That the linear pool of distinct calibrated forecasts is uncalibrated and underconfident is Ranjan and Gneiting (2010), in the real-valued / PIT setting; the “extremize the average” correction (Satopää et al. 2014; Baron et al. 2014) is the operational response. Those statements are qualitative. We give the binary-outcome magnitude: a tight ceiling $CE(\bar{P}) \leq 1/2$ (Prop. 1), a clean closed-form witness $8/27$ with an interpretable Bayesian story (Thm. 2), and the fact that the ceiling is tight (Prop. 3) but only at infinite resolution — making explicit that the “worst-case miscalibration of an ensemble” is not a single number but a function of the reliability diagram’s bin width.

Practical reading. (i) Averaging probabilities is not a calibration-preserving operation, and the damage is not negligible: even two diverse calibrated models can land $8/27 \approx 0.30$ off, and several can approach 0.5. Recalibrate *after* pooling (Platt/isotonic/beta), or extremize. (ii) The resolution caveat doubles as a warning about ECE itself: the metric’s verdict on a pooled forecaster can swing from “perfectly calibrated” to “maximally miscalibrated” purely by changing the bin width, which is exactly the binning sensitivity documented for ECE estimators (Guo et al. 2017; Vaicenavičius et al. 2019; Kumar et al. 2019). A pooled forecaster whose reports cluster near the base rate is the adversarial case where this matters most.

Limitations.

- *Binary outcomes, L^1 error.* Proposition 1’s clean $1/2$ uses $Y \in \{0, 1\}$ and the L^1 (ECE) error. The squared/reliability (L^2) version and multiclass or real-valued Y (where Ranjan–Gneiting live) would have different constants; we did not compute them.
- *$8/27$ is a witness, not the robust maximum.* We did not determine the exact resolution-robust worst case at a fixed bin width b (the value that increases from below toward $1/2$ as $b \rightarrow \infty$). Our numerical search over 2×2 pairs suggests it exceeds 0.45 even at moderate b , but we state no theorem; that constant — and its dependence on b — is left open.
- *$m \rightarrow \infty$ limit.* The m -agent ECE is still rising at $m = 12$ (0.470); we did not compute its limit (a Poisson-regime calculation as $mq \rightarrow \lambda$) and so do not claim it reaches $1/2$.
- *Perfect calibration assumed.* We compare against *perfectly* calibrated inputs to isolate the effect of averaging; with empirically-calibrated inputs the pool inherits their error too.

References

1. R. Ranjan, T. Gneiting. “Combining probability forecasts.” *Journal of the Royal Statistical Society: Series B* 72(1):71–91, 2010. doi:10.1111/j.1467-9868.2009.00726.x.
2. T. Gneiting, F. Balabdaoui, A. E. Raftery. “Probabilistic forecasts, calibration and sharpness.” *Journal of the Royal Statistical Society: Series B* 69(2):243–268, 2007. doi:10.1111/j.1467-9868.2007.00587.x.
3. A. P. Dawid. “The well-calibrated Bayesian.” *Journal of the American Statistical Association* 77(379):605–610, 1982. doi:10.1080/01621459.1982.10477856.
4. V. A. Satopää, J. Baron, D. P. Foster, B. A. Mellers, P. E. Tetlock, L. H. Ungar. “Combining multiple probability predictions using a simple logit model.” *International Journal of Forecasting* 30(2):344–356, 2014. doi:10.1016/j.ijforecast.2013.09.009.
5. J. Baron, B. A. Mellers, P. E. Tetlock, E. Stone, L. H. Ungar. “Two reasons to make aggregated probability forecasts more extreme.” *Decision Analysis* 11(2):133–145, 2014. doi:10.1287/deca.2014.0293.
6. C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger. “On calibration of modern neural networks.” *ICML 2017*. arXiv:1706.04599.
7. J. Vaicenavičius, D. Widmann, C. Andersson, F. Lindsten, J. Roll, T. B. Schön. “Evaluating model calibration in classification.” *AISTATS 2019*. arXiv:1902.06977.
8. A. Kumar, P. Liang, T. Ma. “Verified uncertainty calibration.” *NeurIPS 2019*. arXiv:1909.10155.